POLICY BRIEF • SPRING 2026

Good Intentions Are Not Governance: The Case Against Pfizer's Current AI Framework

Addressed to: Pfizer's Chief Medical Officer, Chief Scientific Officer, and Board-Level R&D Committees

Executive Summary

When Pfizer formally integrated AI into its clinical R&D workflows in 2018, it gained something genuinely powerful: a system capable of processing eligibility data at a scale and speed no human team could match. What it did not gain, for five years, was any binding framework for overseeing that system. By the time the first governance document arrived in 2023, AI had already become the operational infrastructure for patient selection across more than half of Pfizer's clinical trials, making consequential decisions about who gets access to experimental medicine with no audit requirements, no demographic performance thresholds, and no formal Pfizer accountability chain.

Three aspirational principles currently stand in place of that framework. They carry no enforcement mechanisms, no audit requirements, and no accountability chain when outcomes diverge from stated intent. They are a starting point, and Pfizer has not moved past it.

This brief argues that Pfizer must implement four binding governance reforms before any further expansion of AI-assisted patient screening proceeds: mandatory human sign-off on every eligibility decision, demographic bias audits before deployment, a formal AI Ethics Review Board with veto authority, and performance incentives tied to diversity outcomes. Each recommendation is grounded in existing pharmaceutical precedent, current regulatory trajectory, and the financial realities of late-stage trial failure.

I. The Human Cost

A 2014 study by Haynes-Maslow and colleagues conducted focus groups with 82 African American women who had survived cancer or cared for someone who had. Among survivors, 87 percent had never enrolled in a clinical trial, and two-thirds said their physician had never raised the possibility. The barrier was not primarily logistical; participants said they would do whatever it took to get good care. The problem was that nobody had ever framed a trial as part of that care.[1]

That finding reflects a pattern documented across thousands of studies. A 2022 cohort analysis of over 20,000 U.S. clinical trials found Hispanic and Latino patients enrolled at 6 percent, compared with a census share of 16.3 percent; Asian patients appeared in just 1 percent of trials, despite representing approximately 5 percent of the population; and 21 percent of trials enrolled zero Black participants.[2] These figures predate the widespread adoption of AI-assisted screening, which means the systems now accelerating patient selection were built on data that already carried these gaps baked in.

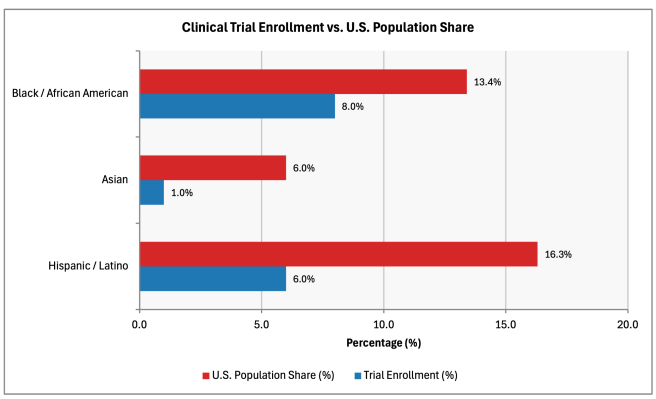

Figure 1: Clinical Trial Enrollment vs. U.S. Population Share by Race/Ethnicity. Original visualization. Readers should note that Hispanic and Latino patients appear at less than half their census share, and 21 percent of trials enrolled zero Black participants, representing gaps that AI systems trained on this data will reproduce and accelerate at scale. Data from Turner et al. (2022).

II. What AI Actually Does Inside a Pfizer Clinical Trial

By 2022, AI was running inside more than half of all Pfizer clinical trials.[3] Three years later, eligibility screening systems were moving through thousands of patient records per hour, work that used to require a human coordinator to review ten or twelve files at a time.[4] The selection pipeline runs through five nodes: protocol design, eligibility screening, site selection, real-world data integration, and quality management. Eligibility screening is where the consequences land most directly, determining who gets flagged and who gets passed over.[5]

Pfizer's 2022 partnership with Truveta added over 50 million patient electronic health records (EHR) from 42 states, refreshed daily, to the data environment those systems draw from.[6] At that volume, the system isn't assisting with eligibility decisions. It's making them.

The mechanism that produces bias here is not complicated. AI models trained on electronic health records learn from patients who most frequently engage with the healthcare system, which systematically underrepresents people in rural areas, low-income communities, and populations with historically limited access to care. Patients who appear less in the data get surfaced less by the model. No deliberate exclusion is required at any point in the process. The discrimination is structurally unintentional, but unfortunately does not make it any less real.

Obermeyer and colleagues documented the clearest version of this in *Science* in 2019. A widely used commercial healthcare algorithm was assigning lower risk scores to Black patients than to white patients with equivalent clinical need because it used healthcare spending as a proxy for health. Black patients were spending roughly $1,800 less annually on care; they were not healthier; rather, they had decades of differential access. The algorithm read that gap as health status information, cutting Black patient identification for high-need care programs by more than half. The manufacturer confirmed the finding.[7]

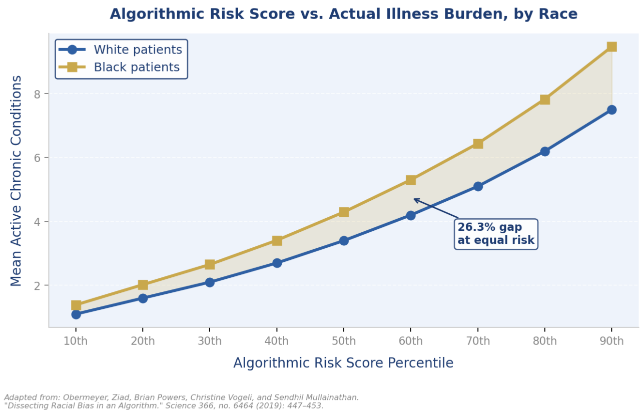

Figure 4: Algorithmic Risk Score vs. Actual Illness Burden, by Race. Adapted from Obermeyer et al. (2019). Readers should observe that at every predicted risk percentile, Black patients carry a 26.3% higher burden of active chronic conditions than White patients with identical risk scores — demonstrating that cost-based proxies systematically underestimate the clinical needs of Black patients. Source: Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan, "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations," Science 366, no. 6464 (2019): 447–453, https://doi.org/10.1126/science.aax2342.

Pfizer's eligibility systems operate on the same category of data under the same structural conditions: EHR-derived records where healthcare utilization proxies for health status, applied at population scale, with no mechanism to flag when that proxy breaks down across demographic groups.

III. The Governance Lag

Table 1: Pfizer AI Deployment vs. Governance Policy Milestones (2018–2025)

The table tracks Pfizer's AI deployment milestones alongside the arrival of governance policy from 2018 to 2025. Readers should note that binding audit requirements remain absent as of 2025, seven years into active AI deployment across clinical trial workflows.

Year	Deployment Milestone	Governance Milestone
2018	AI formally integrated in clinical R&D	—
2020	SDQ (Smart Data Query) tool cuts data cleaning from 30 days to 22 hours	—
2022	Truveta partnership: 50M+ patient EHR dataset	—
Feb. 2023	—	Three Principles published
Dec. 2023	—	AI Policy Position paper
Aug. 2024	—	Chief AI and Analytics Officer appointed
2025	AI processes thousands of eligibility records per hour	No binding audit protocols

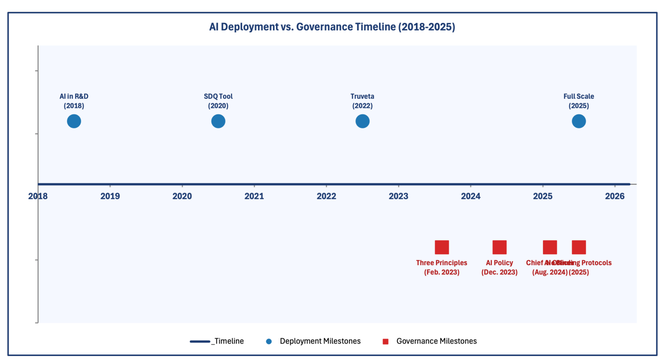

Figure 2: Pfizer AI Deployment vs. Governance Policy Milestones (2018–2025). Original visualization. Readers should observe the five-year gap between AI deployment (2018) and the first governance document (2023), during which AI was making patient selection decisions with no audit requirements, no accountability chain, and no binding oversight of any kind.

The comparison that comes to mind is financial services. When banks started deploying algorithmic decision-making in consequential contexts, the Federal Reserve issued SR 11-7 in 2011, mandating independent model validation, back-testing requirements, board-level accountability, all in place before significant failures had accumulated.[8] Pfizer has had seven years to build equivalent infrastructure. It hasn't done so.

The December 2023 AI Policy Position paper acknowledges that AI systems are "vulnerable to biases, occurring at the point of data collection, the development of algorithms, and finally in the use of the system."[9] What the document doesn't include is any mechanism for actually detecting or correcting what it describes. Mittelstadt's research at Oxford is direct on this point: principle-based AI governance frameworks without enforcement structures are unlikely to produce the behavioral change they're meant to address.[10]

IV. The Financial Case for Getting This Right

The efficiency case for AI in clinical trials is faster screening, lower recruitment costs, and compressed timelines. But what those efficiencies are feeding into matters. Drug development carries a 7.9 percent approval rate from Phase I through regulatory clearance.[11] Phase III trials represent the single largest cost center in drug development,[12] and under DiMasi and colleagues' capitalized-cost methodology, a single approved drug represents $2.558 billion in cumulative development investment.[13] A bias-related failure at late stage means writing off somewhere between seven and seventeen years of that.[14]

AstraZeneca completed a full pharmaceutical AI ethics audit, running from start to finish in one quarter, approximately 2,000 person-hours.[15] At standard industry rates, that represents roughly $300,000–$500,000 in cost: less than 0.02 percent of the pipeline asset it protects. These numbers don't require much interpretation.

The post-approval picture compounds the risk. Between 2014 and 2021, fewer than 20 percent of FDA-approved drugs had trial data on efficacy or side effects specifically for Black patients.[16] Drugs developed under those conditions carry a specific kind of risk: the effective population may be narrower than projected, outcome divergences draw post-market shareholder scrutiny, and in some cases, regulatory withdrawal follows.

The pattern is documented across therapeutic areas. Asthma treatments have demonstrated efficacy gaps in patients of African American and Puerto Rican descent; cardiovascular interventions developed from predominantly male trial populations have produced measurable outcome disparities in women.[17]

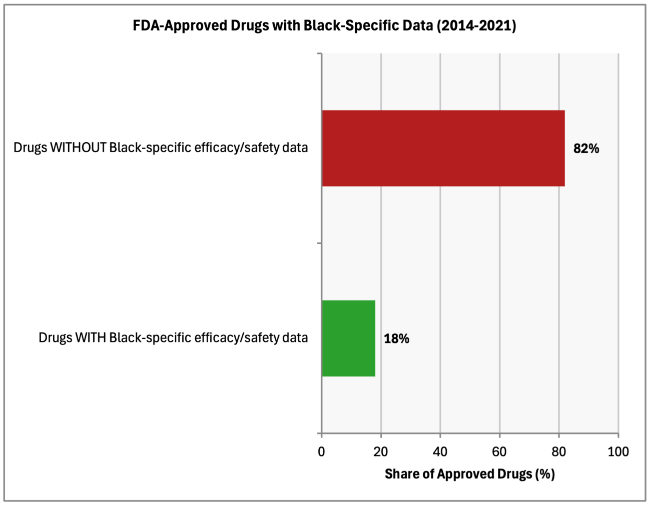

Figure 3: FDA Drug Approval Diversity Gaps (2014–2021). Readers should note that more than 80 percent of FDA-approved drugs in this period carried no efficacy or safety data for Black patients, meaning the post-approval liability and market access risks described above are structural features of the current system, not edge cases. Source: USC Schaeffer Center (2022).

The market access consequences compound the pipeline risk. A drug lacking efficacy data for Black Americans (who represent 13.7 percent of the U.S. population) may receive a narrower label indication, face resistance from payers requiring post-market studies before full reimbursement, or become the subject of litigation as outcome divergences accumulate. Each of these is a revenue consequence.

A drug with a genuinely diverse evidence base reaches a larger market, with fewer post-approval complications, and that advantage compounds across a portfolio. Building governance infrastructure that produces representative trial data is also, in purely practical terms, how Pfizer protects the commercial value of what's in the pipeline.

V. Four Recommendations

The four recommendations below are grounded in existing pharmaceutical and financial services precedent and are implementable within Pfizer's current organizational structure. A pre-deployment bias audit, modeled on the AstraZeneca approach, runs approximately $300,000–$500,000, a one-time cost recoverable within weeks of trial acceleration. An AI Ethics Review Board requires no new headcount beyond designating a standing committee chair and formalizing a review cadence within existing governance infrastructure. Explainability documentation (SHAP/LIME) is integrated directly into model development pipelines at negligible marginal cost. Performance metric alignment requires a policy update, not a budget line.

A. Enforceable Internal Rules

- Every AI-generated patient eligibility decision requires documented human sign-off–the reviewing clinician's name, credentials, and stated rationale, recorded permanently in the trial audit trail.
- No patient should be excluded through a fully automated process without explainability documentation. SHAP (Shapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), the most widely used model interpretation tools in peer-reviewed clinical AI research, allow clinicians to interrogate why the model produced a given output before it's acted upon.[18]
- Demographic performance audits are required before any model transitions from internal development to global deployment, with equalized odds and demographic parity (two standard fairness metrics that measure whether a model's error rates are consistent across racial and demographic groups, as the operative benchmarks.[19]
- A formal AI Ethics Review Board (clinical scientists, biostatisticians, patient advocates) holds genuine veto authority over deployment decisions and reports directly to the Chief Medical Officer.

B. AI Literacy for Clinical Teams

- R&D teams need training that equips them to critically assess algorithmic outputs.
- SHAP and LIME belong in standard trial preparation workflows.

C. System Restructuring

- Mandatory bias validation checkpoints at every stage of development, with board-level sign-off before any model deploys at scale.
- Standardized documentation for every patient selection model: architecture, training data composition, validation cohort demographics, and ongoing subgroup performance.
- Proactive alignment with the FDA's January 2025 draft guidance and the European Medicines Agency (EMA)'s September 2024 Reflection Paper, both of which require documentation that Pfizer's current governance structure can't produce.[20]

D. Incentivize Proper Governance

- Diversity documentation quality becomes a formal component of R&D team performance metrics.

- FDA’s Quality Management Maturity program links proactive governance to reduced inspection burden and faster submissions.[21] Companies that build this ahead of the mandate consistently get better treatment in the review process.

VI. Addressing Objections

“This is the FDA’s job.”

FDA’s January 2025 guidance is still in draft. That makes internal governance the only patient protection mechanism currently in operation. The relationship between internal governance quality and regulatory treatment runs consistently in one direction: companies demonstrating proactive governance get reduced inspection burden and faster submission pathways.[22] Building the infrastructure before it’s mandated also means Pfizer has a voice in how the mandate gets written.

“This will slow down our trials.”

Drugs receiving FDA Breakthrough Therapy designation reached approval in a median of 4.8 years, compared to 8.0 years for non-designated drugs.[23] What that demonstrates is that intensive regulatory engagement earlier in development reduces the friction that causes late-stage delays, not the other way around. A governance structure that catches demographic bias before Phase III works the same way. And a failed Phase III trial, with everything it erases, is the actual cost of the current approach.

“We don’t have the budget for this.”

A pre-deployment bias audit modeled on the AstraZeneca approach costs approximately $300,000–$500,000, which is less than 0.02 percent of the capitalized investment in a single pipeline asset. An AI Ethics Review Board requires no new headcount, only a standing mandate and meeting cadence within existing governance infrastructure. SHAP and LIME explainability tools are open-source and integrate into existing model development pipelines. The full governance framework described here is implementable within a single fiscal quarter at a cost that rounds to zero against the pipeline value it protects. The question is not whether Pfizer can afford it. The question is whether it can afford a bias-driven Phase III failure that it could have prevented.

“The Three Principles already cover this.”

Roche’s Data Ethics Principles explicitly name monitoring, governance, and disciplinary actions as operationalized components of its AI framework; it’s the only published pharmaceutical AI governance document in the sector that specifies enforcement consequences. Novartis runs a four-tier AI risk classification system aligned with the EU AI Act, overseen by a senior executive committee with authority over all deployment decisions.[24] Both companies have built enforcement infrastructure around their stated commitments. The standard Pfizer is behind isn’t theoretical. It already exists, and the companies Pfizer competes with are maintaining it.

VII. Conclusion

Pfizer built exactly the AI infrastructure a modern pharmaceutical company should have, and the scientific case for that investment is sound. What the deployment timeline makes clear is that the pipeline was optimized for speed and scale without the checkpoints patient safety requires at that speed and scale.

The patients most likely to be missed by the current system are the same patients for whom an approved drug may not perform as expected, because they were never in the data that produced the approval. Congress recognized this directly, enacting diversity action plan requirements for Phase III trials as part of the Food and Drug Omnibus Reform Act of 2022.[25] FDA is now moving toward requiring documentation of it. And Pfizer’s competitors are already building toward the standard that documentation will require.

The question for this institution's leadership is whether that work happens on Pfizer's terms, or in response to a requirement it could have anticipated.

This is also a question of public trust. Clinical trial participation depends on communities believing that pharmaceutical companies are selecting patients fairly. When algorithmic bias is exposed, as it has been repeatedly in adjacent healthcare contexts, the reputational damage extends beyond the company to the trials themselves, reducing enrollment precisely among the populations most underrepresented. A governance framework that Pfizer can demonstrate publicly is not just a regulatory asset; it is the condition under which broad participation remains possible.

Footnotes

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